



Eikonix: A Digital E- Commerce Space for Traditional and Digital Artworks

A System Integration and Architecture Project Documentation
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DEDICATION

This project is dedicated to our families, friends, instructors, and classmates who continuously supported and motivated us throughout the development of this study. Their encouragement, guidance, and understanding inspired the researchers to complete this project despite the challenges encountered during the system development process.

We also dedicate this project to future students, artists, and aspiring developers who may benefit from the ideas, innovations, and experiences presented in this study. May this project serve as inspiration for the continuous improvement of digital platforms that support creativity, innovation, and technology.



ACKNOWLEDGEMENT

The researchers would like to express their sincere gratitude to everyone who contributed to the completion of this project entitled “Eikonix: A Digital E-Commerce Space for Traditional and Digital Artworks.”

First and foremost, the researchers would like to thank our instructors for their guidance, valuable suggestions, and continuous support throughout the development of the system and documentation process. Their knowledge and expertise greatly contributed to the improvement of the study.

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Special thanks are given to our families for their unwavering emotional, moral, and financial support. Their patience and encouragement motivated the researchers to continue and complete this study.

Lastly, the researchers express their gratitude to Almighty God for providing wisdom, strength, and perseverance throughout the completion of this project.



EXECUTIVE SUMMARY

Eikonix: A Digital E-Commerce Space for Traditional and Digital Artworks is an integrated web-based e-commerce platform developed to provide artists and customers with a centralized and secure environment for buying and selling artworks online. The system supports both traditional and digital artworks while offering user-friendly functionalities for customers, administrators, and suppliers.

The enhanced version of the project introduces an Inventory Management Integration module that improves product monitoring, stock handling, and supplier coordination. A new “Supplier” user role was integrated into the database to support inventory operations. Suppliers are given limited system access focused solely on product management and inventory-related functions. This integration enhances the overall workflow efficiency between administrators and suppliers while maintaining proper access control.

The project also improves the existing Admin Dashboard by refining inventory handling and product visibility. Products with zero stock are automatically removed from the customer products table to prevent unavailable purchases. In the Admin Dashboard, these products are labeled as “Sold”, while they remain visible in the sales reports and analytics section for historical tracking and reporting purposes. Additional validation logic was also implemented to prevent duplicate artwork names from being added into the system database.



Several user interface and user experience improvements were also applied. Theme toggling issues affecting some landing page sections were fixed, layout inconsistencies in the featured cards section were corrected, and the inventory dashboard interface was redesigned into a simplified single-page scrollable layout for better usability.

Overall, Eikonix aims to provide a modern, accessible, and organized digital marketplace that supports artists, simplifies inventory management, and improves the online shopping experience for customers.



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CHAPTER I

PROJECT BACKGROUND

Introduction

The continuous advancement of information technology has significantly transformed modern business operations, particularly in the field of e-commerce. Businesses and independent sellers now rely on digital platforms to promote products, manage transactions, and expand market reach efficiently. According to Wulfert et al. (2022), e-commerce ecosystems help facilitate coordination, transactions, and value co-creation among different users through integrated digital platforms.

In recent years, the art industry has also adapted to digital transformation by utilizing online marketplaces and digital platforms to promote and sell artworks. According to Lee (2022), digital art platforms provide emerging artists with opportunities to gain visibility, connect with audiences, and establish legitimacy within the online art market. These platforms have become increasingly important because traditional methods of artwork promotion are often limited by location, accessibility, and operational costs.

The growth of online commerce also increased the need for effective inventory management and organized transaction systems. Supported by Ngolu et al. (2022), electronic inventory systems improve stock monitoring, operational efficiency, and product management in e-commerce platforms. Proper inventory management also helps reduce transaction errors and improves customer satisfaction.



Eikonix was developed as a web-based e-commerce platform specifically designed for traditional and digital artworks. The system provides artists and customers with a centralized online marketplace where artworks can be displayed, managed, and purchased conveniently. The platform includes features such as product browsing, shopping cart management, order processing, inventory monitoring, and sales reporting.

The enhanced version of the system integrates an Inventory Management module that improves stock monitoring and product organization. A supplier role was also added to support inventory-related operations while maintaining role-based access control. Additional improvements include duplicate artwork validation, automatic removal of sold-out products from the products page, and user interface optimizations for better usability and responsiveness.

Through these enhancements, Eikonix aims to provide a more organized, accessible, and efficient digital marketplace for artists, administrators, suppliers, and customers.

Project Background and Context

The continuous growth of e-commerce platforms has changed the way products and services are marketed and sold. Online marketplaces provide businesses with wider customer reach, improved accessibility, and more efficient transaction management. According to Engert et al. (2022), digital platforms support collaboration and interaction among different stakeholders through integrated online ecosystems.



The art industry has also experienced significant digital transformation in recent years. According to Lee and Lee (2022), digital platforms play an important role in helping artists promote and sell their artworks online. Additionally, the increasing popularity of digital artworks and online art marketplaces demonstrates the growing dependence of the creative industry on digital commerce.

During the initial phase of the Eikonix project, the researchers developed a web-based e-commerce platform that allowed customers to browse and purchase traditional and digital artworks online. The earlier version of the system included core functionalities such as customer registration, product management, shopping cart operations, order management, customer monitoring, and sales reporting.

After evaluating the previous version of the system, the researchers identified the need to improve inventory monitoring and product organization. According to Ngolu et al. (2022), inventory management systems improve stock control and operational efficiency in e-commerce environments. Because of this, the enhanced version of Eikonix integrates an Inventory Management Dashboard that supports stock monitoring and supplier coordination.

Additional improvements were also implemented in the enhanced system. Validation logic was added to prevent duplicate artwork names from being entered into the database. Sold-out products are automatically removed from the products page while remaining visible in sales reports for historical monitoring purposes. User interface issues related to



theme toggling and layout responsiveness were also corrected to improve the overall user experience.

Through these integrations and improvements, Eikonix evolved into a more organized and scalable e-commerce platform capable of supporting efficient artwork transactions, inventory management, and product monitoring.

Project Purpose and Description

The purpose of this project is to develop an integrated web-based e-commerce platform that provides artists with a centralized space to sell and manage traditional and digital artworks online. The system aims to simplify artwork transactions while improving inventory management, product monitoring, customer interaction, and sales reporting.

Eikonix allows customers to browse available artworks, add products to their cart, place orders, and manage purchases through an accessible digital platform. Administrators are provided with management tools for monitoring products, orders, customers, and sales analytics.

The improved version of the project introduces an Inventory Management Integration feature that enhances product stock handling and coordination between administrators and suppliers. A dedicated supplier role was implemented to manage products efficiently without granting unnecessary administrative permissions.



Main Objective

The main objective of this study is to develop and enhance Eikonix, a web-based e-commerce platform for traditional and digital artworks, by integrating inventory management functionalities, improving product and stock monitoring, strengthening data validation processes, and enhancing user experience through optimized system interfaces and organized access control for administrators, suppliers, artists, and customers.

Specific Objectives

1. To develop a web-based e-commerce platform for displaying and selling traditional and digital artworks.
2. To integrate an inventory management module that improves stock monitoring and product management.
3. To implement a supplier user role with limited access dedicated to product and inventory handling.
4. To improve product management by preventing duplicate artwork entries through validation logic.
5. To enhance customer experience by automatically hiding sold-out products from the products page while retaining them in sales reports.
6. To improve the usability and responsiveness of the system interface through layout and theme-related refinements.
7. To provide administrators with organized dashboards for monitoring products, orders, customers, inventory, and sales performance.



Scope and Delimitations of the Project

Scope

The system is capable of performing the following functions:

1. Allows users to register and create customer accounts.
2. Allows customers to browse traditional and digital artworks.
3. Enables customers to search, add products to cart, and place orders.
4. Allows administrators to manage products, orders, customers, and sales reports.
5. Provides inventory monitoring and stock management functionalities.
6. Allows suppliers to manage products and inventory through a dedicated inventory dashboard.
7. Prevents duplicate artwork names through validation checking.
8. Automatically removes sold-out products from the customer products listing.
9. Retains sold-out products in sales reports and analytics for historical monitoring.
10. Supports dark mode and light mode interface customization.
11. Generates sales summaries and order reports for administrative monitoring.
12. Provides role-based access control for customers, suppliers, and administrators.



13. Allows administrators to activate or deactivate customer accounts.

14. Supports responsive and organized user interface layouts.

Delimitations

The system has the following limitations:

1. The platform is limited only to artwork-related products.
2. The system does not support real-time shipment tracking.
3. The platform does not include integrated online payment gateway APIs.
4. Suppliers are limited only to inventory and product management functions.
5. The system does not support mobile application deployment and is limited to web-based access.
6. The system does not include advanced artificial intelligence recommendation features.
7. The platform cannot automatically generate physical shipping labels.
8. The platform requires internet access to operate properly.
9. The project focuses only on small-to-medium scale e-commerce operations for artworks.
10. The system does not support cloud-based media storage; all artwork images are stored in a local folder on the same machine as the system.
11. The system does not automatically apply watermarks or copyright tags (e.g., “Eikonix” labels) to uploaded artwork images.



12. The system does not include batch uploading or bulk import of artwork listings.
13. The system does not include an automated backup or recovery mechanism for the database and media files.
14. The system does not process real payments. The checkout flow is implemented for order simulation purposes.

Operational Definition of Terms

Administrator. A person responsible for managing the overall operations of the system. In this study, the administrator manages products, customers, orders, inventory records, and sales reports.

Artwork. A creative visual product produced by an artist. In this study, artworks refer to traditional and digital art products sold within the Eikonix platform.

Customer. A person who purchases products or services from a business. In this study, customers are registered users who browse, order, and purchase artworks through the system.

Dashboard. A graphical interface that displays important information and controls. In this study, dashboards are used by administrators and suppliers to monitor products, inventory, and reports.

Digital Artwork. Art created using digital technology or software applications. In this study, digital artworks are products available for online purchase through the platform.



E-Commerce. The process of buying and selling products or services through electronic systems such as the internet. In this study, e-commerce refers to the online transaction functionalities of Eikonix.

Eikonix. The proposed web-based e-commerce platform developed for selling and managing traditional and digital artworks.

Inventory Management. The process of monitoring and controlling product stock levels. In this study, inventory management refers to the integrated product stock monitoring and supplier coordination features of the system.

Product Management. The process of organizing, updating, and monitoring product information. In this study, product management includes adding, editing, deleting, and validating artworks within the platform.

Sales Report. A summary of sales transactions and performance data. In this study, sales reports display total sales, orders, customers, and product-related analytics.

Supplier. A person or role responsible for handling product inventory and stock information. In this study, suppliers are limited-access users who manage products through the inventory dashboard.

System Integration. The process of combining multiple system components into a unified platform. In this study, integration refers to the addition of inventory management functionalities into the existing Eikonix system.



Traditional Artwork. Physical or hand-made artwork created using traditional materials and techniques. In this study, traditional artworks are listed and sold through the Eikonix platform.

User Role. A defined access level assigned to users within a system. In this study, user roles include administrator, supplier, and customer accounts.

Validation Logic. A programming process used to verify data accuracy and consistency. In this study, validation logic is used to prevent duplicate artwork names from being added to the system.



CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

Foreign Literature, Studies and Systems

E-commerce Platform

E-commerce systems have significantly transformed business operations by enabling online transactions, wider market reach, and efficient product management. These platforms allow businesses to manage products, customers, and transactions through integrated digital systems.

According to Wulfert et al. (2022), e-commerce ecosystems improve coordination and value creation among users through digital platforms. Likewise, Laudon and Traver (2023) stated that e-commerce systems enhance business efficiency by expanding customer reach and supporting digital transactions. Turban et al. (2018) further explained that e-commerce platforms create digital marketplaces that connect buyers and sellers through secure and structured online environments. These findings support the use of web-based platforms like Eikonix for managing online artwork transactions.

Digital Art Marketplace

Digital platforms have changed how artworks are promoted and sold by increasing accessibility and connecting artists to wider audiences. Fernandes and Afonso (2020)



found that online art platforms improve transparency and communication between artists and buyers while reducing reliance on physical galleries. Similarly, Velthuis and Coslor (2021) noted that digital marketplaces increase artist visibility and market opportunities. Kaplan and Haenlein (2010) emphasized that digital platforms and online communities allow creators to reach global audiences, supporting the growth of online creative industries such as digital art marketplaces. These studies highlight the importance of web-based platforms in supporting modern art commerce.

Inventory Management System

Inventory systems are essential for improving stock control, accuracy, and operational efficiency in e-commerce platforms. Ramos-Miller and Pacheco (2023) showed that web-based inventory systems with dashboards, automated reporting, and role-based access improve efficiency and data accuracy. Laudon and Laudon (2020) emphasized that management information systems help organizations organize data, monitor operations, and improve decision-making processes, particularly in inventory and business management systems. O'Brien and Marakas (2011) also explained that information systems enhance organizational efficiency by supporting real-time data processing and structured business operations. These features support better product and inventory management in online systems like Eikonix.



Local Literature, Studies and Systems

Digitalization of Traditional Arts

Digital technology helps preserve and promote traditional artworks through online platforms and virtual exhibitions. The study *The Impact of Applied Informatics and Digitalization of Traditional Arts* (2024) found that digitalization expands audience reach and provides more opportunities for artists to sell and promote their works online. These findings support the integration of traditional and digital artworks in web-based platforms like Eikonix. In addition, the National Commission for Culture and the Arts (NCCA, 2022) emphasized that digital platforms play a vital role in preserving Philippine cultural heritage while increasing accessibility through online exhibitions and digital archives.

Online Platform for Filipino Artists

Online art marketplaces are increasingly important for Filipino artists in promoting and selling their works. The study *i-Sining: An Online Platform for Filipino Artists* (2023) emphasized the need for dedicated digital platforms to support local artists. It highlighted that structured online systems improve accessibility and help sustain artists in the digital market. Similarly, Bayudan-Dacuycuy and Sinsay (2022) found that Filipino artists and small creative entrepreneurs benefit from digital platforms, although challenges such as



infrastructure limitations and system inefficiencies still exist, highlighting the need for improved platforms like Eikonix.

Inventory Management System

Inventory systems are essential for improving stock control, accuracy, and operational efficiency in e-commerce platforms. Delin et al. (2025) found that integrating inventory tracking and product management improves workflow efficiency and customer satisfaction in e-commerce systems. Similarly, Bernasol et al. (2021) emphasized that web-based inventory systems with dashboards, supplier profiling, and automated monitoring improve stock accuracy and operational efficiency. In addition, De Guzman and Rivera (2023) stated that web-based systems with inventory and transaction modules significantly improve product organization, order processing, and customer management in Philippine small businesses. These features support better product and inventory management in online systems like Eikonix.

Synthesis of the Reviewed Literature, Studies and Systems

The continuous growth of e-commerce has transformed business operations by enabling online transactions, wider market reach, and efficient product management. Studies by Wulfert et al. (2022) and Laudon and Traver (2023) emphasized that e-commerce



ecosystems and systems improve coordination, transactions, and value creation among users through integrated digital platforms. Turban et al. (2018) further explained that e-commerce platforms create structured digital marketplaces that securely connect buyers and sellers. These findings support the use of web-based platforms in managing products, customers, and business operations efficiently. The art industry has also adapted to digital transformation through online marketplaces and digital platforms. Fernandes and Afonso (2020) found that online art platforms improve accessibility, transparency, and communication between artists and buyers while reducing dependence on physical galleries. Similarly, Velthuis and Coslor (2021) highlighted that digital marketplaces increase artist visibility and market opportunities. Kaplan and Haenlein (2010) emphasized that digital platforms and online communities enable creators to reach global audiences, strengthening the growth of online creative industries such as digital art marketplaces. In the Philippine context, *The Impact of Applied Informatics and Digitalization of Traditional Arts* (2024) and the National Commission for Culture and the Arts (NCCA, 2022) both highlight that digital technology helps preserve and promote traditional artworks while expanding access through virtual exhibitions and digital archives. Likewise, *i-Sining: An Online Platform for Filipino Artists* (2023) and Bayudan-Dacuycuy and Sinsay (2022) emphasize the importance of structured digital platforms in supporting Filipino artists by improving accessibility, visibility, and participation in online art markets.



Inventory management was also identified as an essential feature of e-commerce systems. Ramos-Miller and Pacheco (2023) demonstrated that web-based inventory systems with dashboards, automated reporting, and role-based access improve efficiency and data accuracy. Laudon and Laudon (2020) explained that management information systems support organizations in organizing data, monitoring operations, and improving decision-making processes. O'Brien and Marakas (2011) further emphasized that information systems enhance organizational efficiency through real-time data processing and structured business operations. In addition, Delin et al. (2025), Bernasol et al. (2021), and De Guzman and Rivera (2023) all found that inventory tracking, supplier coordination, and transaction monitoring improve product organization, workflow efficiency, and customer satisfaction in online systems.

Overall, the reviewed literature shows that e-commerce platforms, digital art marketplaces, and inventory management systems significantly improve accessibility, operational efficiency, and transaction management. These studies collectively support the development of Eikonix as a web-based platform that provides organized artwork selling, inventory monitoring, supplier coordination, and efficient online transaction management for artists, customers, administrators, and suppliers.



CHAPTER III

TECHNICAL SPECIFICATIONS

This chapter discusses the technical specifications of the hardware and software requirements that have been used in developing and implementation of this project.

SOFTWARE

Category	Software
Operating System	Windows 10 / Windows 11
Web Server	ISS Express
Database	MySQL Server
Integrated Development Environment (IDE)	Microsoft Visual Studio 2022 / 2026
Framework	ASP.NET MVC 5
.NET version	.NET Framework 4.7.2
Object Relational Mapping	Entity Framework 6 (EF6)
Database GUI Tool	MySQL Workbench CE
Version Control	Git and Github
Web Browser	Google Chrome, Brave, Mozilla Firefox

Table 1 Software Requirements Specification



HARDWARE

Component	Minimum Specifications
Processor	Intel core i3 (or equivalent)
RAM	4 GB
Storage	100 GB HDD
Network	LAN / Wi-Fi connection
Server Computer	Windows OS with IIS installed
Client Computer	Any PC with modern web browser

Table 2 Hardware Requirements Specification

TECHNOLOGIES

Technology	Purpose
ASP.NET MVC 5	Server-Side web framework following Model-View-Controller Architecture
.NET Framework 4.7.2	Runtime environment for the web application
Entity Framework 6 (EF6)	ORM for database operations
MySQL	Relational Database for storing users, products, and orders.
jQuery	DOM manipulation and AJAX requests
JavaScript	Client-side interactivity
IIS Express	Web server hosting the ASP.NET web application
HTML5 / CSS3	Frontend structure and styling

Table 3 Technologies Requirements Specification



DIAGRAMS AND FLOWCHARTS

The comprehensive suite of flowcharts, architectural diagrams, and network layouts maps out the entire operational, structural, and behavioral ecosystem of the Eikonix platform. By defining user paths through the customer, admin, and supplier flowcharts alongside the system inventory logic, the documentation explicitly reinforces the single-stock business rule where purchased unique artworks are instantly hidden from the catalog while remaining preserved for analytics. Furthermore, the context diagram, use case diagram, and network layout bridge the gap between process and infrastructure, visually demonstrating how different user roles securely interact with system boundaries, backend logic engines, and isolated network zones.

FLOWCHART

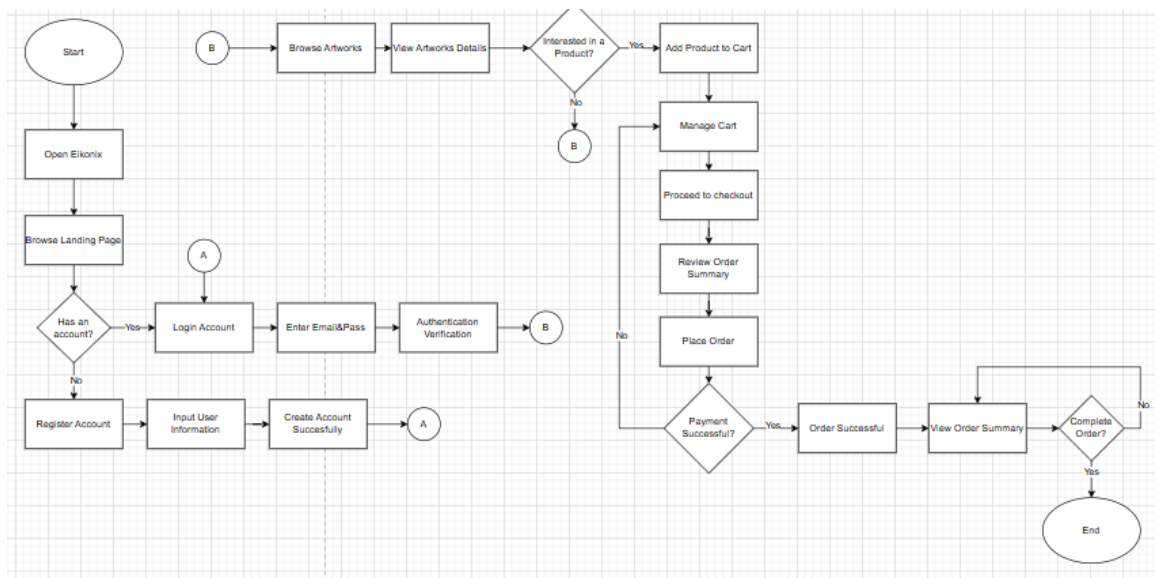


Figure 1.0 Customer Flowchart

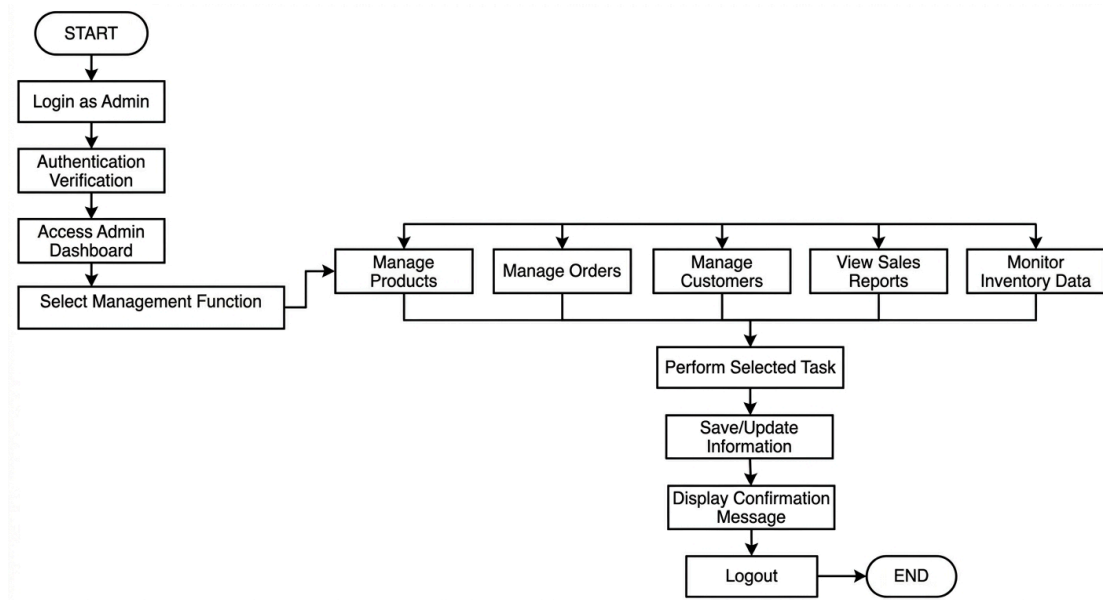


Figure 1.1 Admin Flowchart

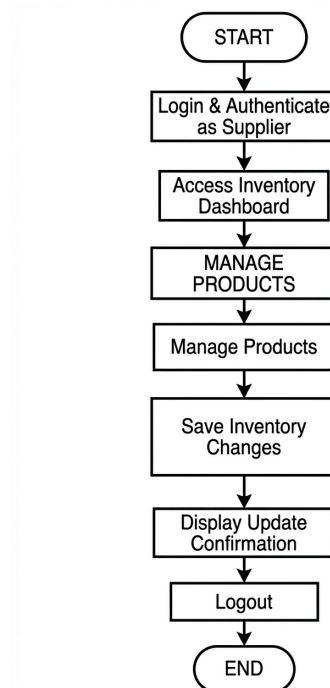


Figure 1.2 Supplier Flowchart

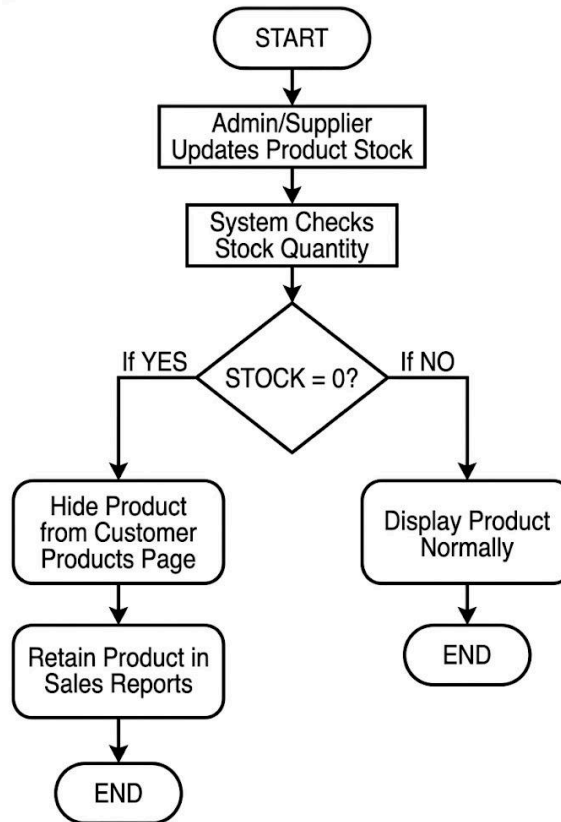


Figure 2 System Inventory Logic Flow

CONTEXT DIAGRAMS

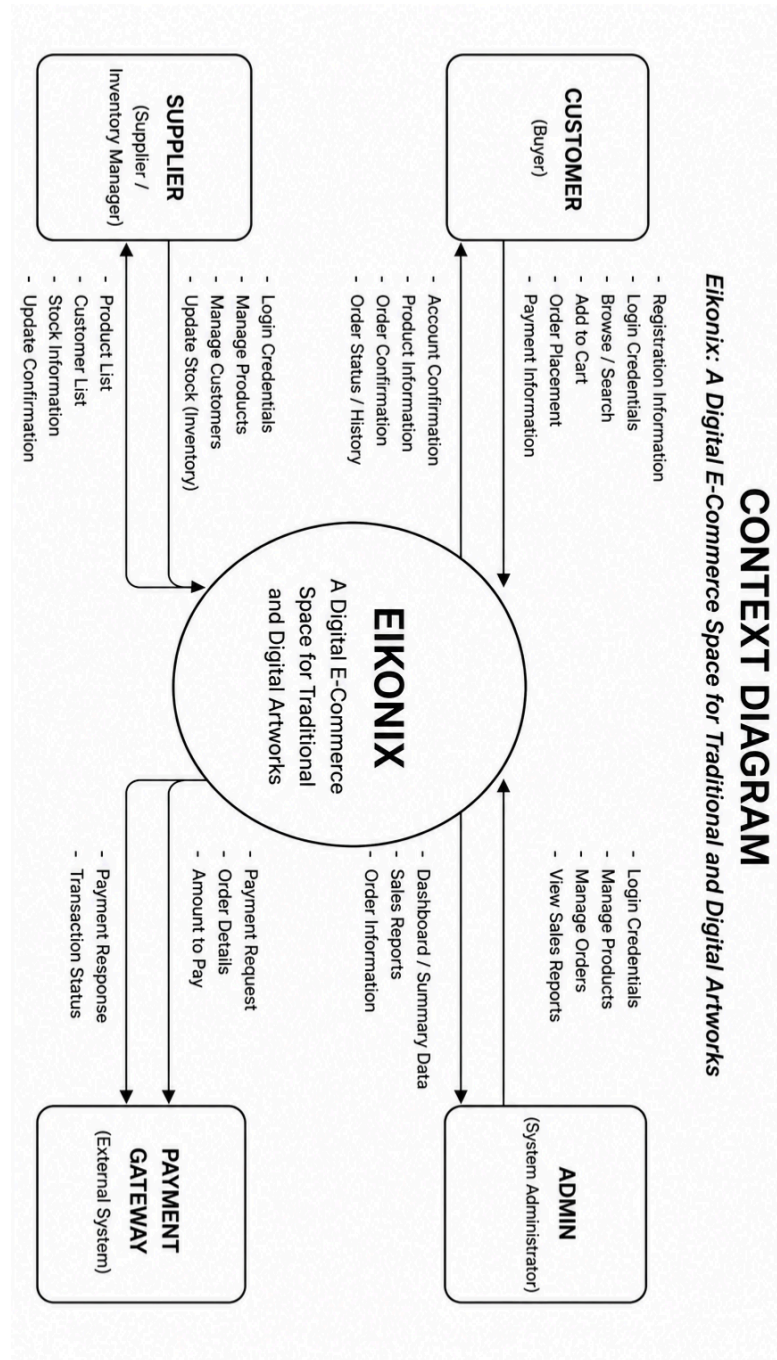


Figure 3 Context Diagram

USE CASE DIAGRAM

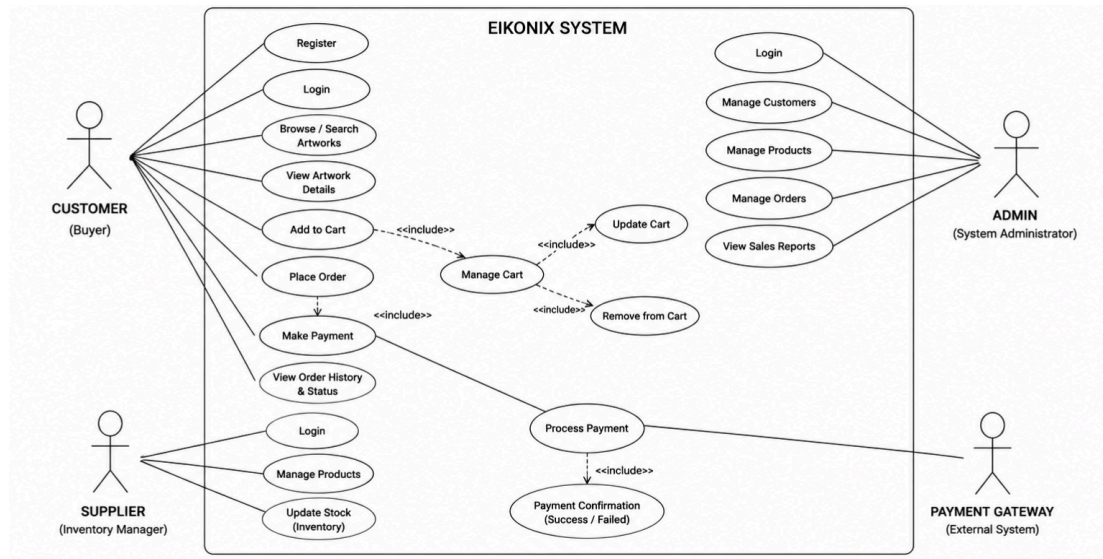


Figure 4 Use Case Diagram

NETWORK LAYOUT

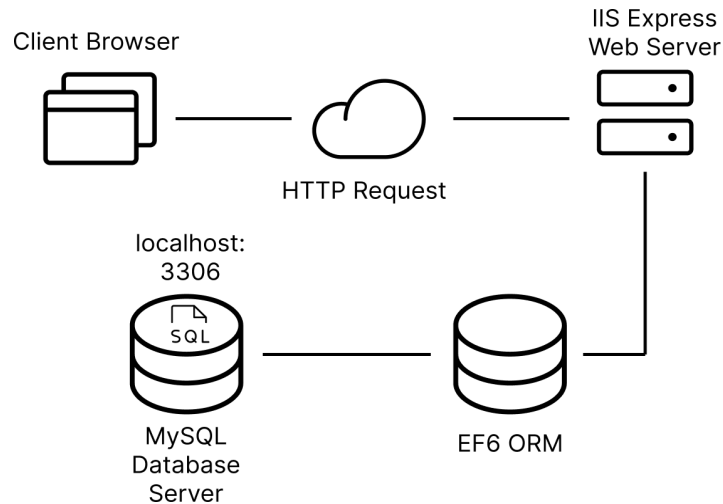


Figure 5 Network Layout

The network layout diagram illustrates how the Eikonix system operates using a client-server architecture. Users access the platform through a web browser, which sends HTTP requests to the IIS Express Web Server. The server processes these requests and communicates with the MySQL Database Server through the EF6 ORM framework to retrieve, store, and manage system data such as user accounts, products, orders, and inventory records. This setup ensures organized communication, efficient data processing, and centralized database management for the Eikonix platform.



CHAPTER IV

METHODOLOGIES

This chapter describes the methodologies used by the proponents in developing the project.

System Requirement Specifications

A. Functional Requirements

1. User Authentication and Role Management

The system shall support three user roles: Customer, Admin, and the newly integrated Supplier role. Each role shall have distinct access permissions.

2. Supplier Inventory Management

The system shall allow users with the Supplier role to add, edit, delete, and update product stock through a dedicated inventory dashboard. Suppliers shall not have access to order management, customer data, or sales analytics.

3. Duplicate Product Name Validation

The system shall prevent the creation of products with the duplicate artwork names and display an appropriate error message.

4. Stock-Based Product Visibility

The system shall hide products with zero stock from the products table and from the supplier inventory dashboard. Zero-stock products shall remain in the admin



dashboard's manage products table where they are displayed with a "Sold" label, sales report and insights.

5. Admin Dashboard Integration

The system shall display key metrics including Total Artworks, Total Orders, Total Customers, and Total Sales. The previously implemented performance metrics (Fair, Excellent, Good) shall no longer appear.

6. Order Management

The system shall allow admins to update order statuses (Pending, Processing, Completed) and cancel orders when necessary.

7. Customer Management

The system shall allow admins to view customer details and deactivate customer accounts.

8. Sales Reporting

The system shall generate sales reports, including Total Sales, Total Orders, Average Order Value, Monthly Sales Summary, Sold Artworks, and Category Performance.

9. Frontend User Features



The system shall provide product browsing, search functionality, add to cart, checkout, and theme toggling (Dark/Light Mode) with consistent application across all pages and sections.

B. Non-Functional Requirements

1. Usability

The system shall provide an intuitive interface for all roles. The supplier dashboard shall be single screen without a complex sidebar for ease of use.

2. Security

Passwords shall be hashed before storage. Role-based access control shall strictly enforce that suppliers and customers cannot access admin-only pages.

3. Performance

Product listings and search results shall load within seconds under normal network conditions.

4. Reliability

The duplicate artwork name shall trigger immediately upon submission to prevent data inconsistency.

5. Scalability

The database design shall support additional user roles and product categories without major restructuring.



System Development Methodologies

The researchers used an iterative development methodology for the development of Eikonix. This approach was selected because the project involved integrating new features into an existing system from IPT102 rather than building from scratch. An iterative methodology allowed the team to work in flexible cycles, where features were added, tested, and refined incrementally based on evolving requirements.

The development process did not follow a rigid linear flow. Instead, the researchers worked in informal but continuous cycles. Each cycle focused on specific integration tasks, such as adding the supplier role, implementing duplicate artwork name validation, modifying stock visibility rules, or refining frontend designs. After completing a task, the researchers tested the change to see how it behaved, gathered feedback through group discussions, and proceeded to the next task. This allowed the researchers to backtrack or add new features when necessary.

Reusing old code played a key role in the methodology. Eikonix follows an MVC (Model-View-Controller) architecture; the researchers were able to adapt existing admin modules for the supplier inventory dashboard. Views, controllers, and database logic that were already working in the admin panel were copied and modified to suit the supplier role.

Conceptual Framework

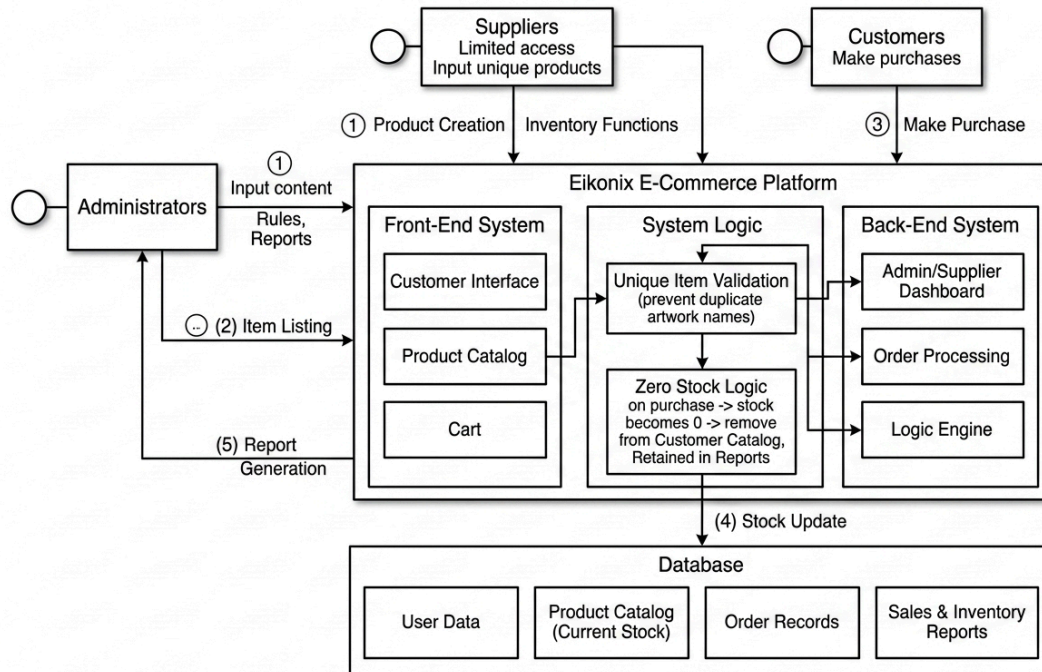


Figure 6 Conceptual Framework

The Conceptual Framework for the Eikonix platform illustrates the interconnected architecture and data flow between the system's primary users including Administrators, Suppliers, Customers and its core logic engines. The architecture maps out how user interactions, such as unique product listings uploaded by suppliers or administrators, are verified by validation logic to prevent duplicate artwork names before being rendered on the front-end product catalog. Once a customer places an order, the back-end system executes the zero-stock logic engine, because the platform deals exclusively in single, unique traditional and digital artworks, the purchase immediately updates the inventory status to zero. This automatically triggers a dynamic database update that hides the sold



artwork from the active customer interface to prevent duplicate purchases, while seamlessly retaining the record within the database's sales and inventory datasets for historical tracking and reporting.

Testing Procedure

A. Unit Testing

Each module of the system was tested to verify that it functions correctly. The following modules were tested:

- User registration and login.
- Role-based access (Customer, Admin, Supplier).
- Product creation, editing, and deletion.
- Shopping cart and checkout.
- Sales report generation.
- Theme toggling (Dark/Light mode).

B. Integration Testing

Integration testing was performed to verify that the new changes and features work correctly with the existing Eikonix system from IPT102. The following test cases were executed:

Feature Tested	Test Scenario	Expected Result	Result
Supplier Dashboard	Supplier logs in and views the inventory dashboard	Dashboard shows: Total products, sold artworks, and a product table	Pass



		with view, edit, delete options. No orders, customers, or other tabs	
Supplier - Create Product	Supplier fills out the form and saves a new artwork	New artwork appears in the product table	Pass
Supplier - View Product Details	Supplier clicks the view button	All artwork details are displayed correctly (Name, artist, description, category, stock, price, medium, software used)	Pass
Supplier - Delete Product	Supplier deletes a product	Product is removed from the table and no longer appears.	Pass
Duplicate Artwork Name	Supplier attempts to create a product with an artwork name that already exists	System rejects submission and displays an error message	Pass
Stock = 0 Visibility	Product gets sold (Order status completed)	Product is hidden from the supplier inventory products table. Product remains visible in the Admin's manage products table and sales reports, with stock displayed as "Sold" or 0 with sold label	Pass

Table 4 Integration Testing



CHAPTER V

SUMMARY AND RECOMMENDATIONS

This chapter contains the summary and recommendations that have been reached by the proponents whilst undertaking this project. The results of the project are summarized to show how effective the system is and recommendations are given as well to further improve future development of the project or for future researchers.

Summary

This study focused on the development and enhancement of Eikonix: A Digital E-Commerce Space for Traditional and Digital Artworks, a web-based platform designed to provide artists and customers with a centralized, organized, and accessible online marketplace for artwork transactions. The system was developed to support both traditional and digital artworks while improving the efficiency of product management, customer interaction, and online selling processes.

The enhanced version of the system introduced an Inventory Management Integration Module that improved stock monitoring and product organization within the platform. A new Supplier user role was implemented to support inventory-related operations while maintaining proper role-based access control. Suppliers were limited to product and stock management functionalities to ensure system security and organized workflow management.



Several improvements were also integrated into the system to enhance usability and operational efficiency. These included the implementation of duplicate artwork validation logic, automatic removal of sold-out products from the customer products page, retention of sold-out products in sales reports for historical monitoring, and refinements to the user interface for improved responsiveness and navigation.

The study demonstrated that integrating inventory management functionalities into the Eikonix platform significantly improved product monitoring, workflow coordination, and system organization. The project also highlighted the importance of user-friendly interfaces, proper access control, and organized inventory handling in the development of modern e-commerce systems.

Overall, the researchers concluded that Eikonix successfully achieved its objective of providing a functional and organized digital marketplace that supports artists, administrators, suppliers, and customers through secure and efficient online transactions.

Recommendations

Based on the findings and limitations of the study, the researchers recommend several improvements and future enhancements for the Eikonix system.

First, future developers may integrate a secure and fully functional online payment gateway API to support real-time electronic payment transactions. This enhancement can improve transaction convenience and provide a smoother purchasing experience for customers.



Second, the researchers recommend the implementation of a real-time delivery and shipment tracking system to allow customers to monitor the status of their orders more efficiently. This feature may improve customer satisfaction and transaction transparency.

Third, future versions of the system may include advanced analytics and reporting tools to provide administrators with deeper insights regarding customer behavior, sales trends, and product performance.

Fourth, the system may be enhanced by developing a dedicated mobile application version to improve accessibility and convenience for users who primarily access online platforms through smartphones and tablets.

Fifth, the researchers recommend integrating artificial intelligence-based recommendation features that can suggest artworks to customers based on their preferences, browsing history, and purchasing behavior.

Lastly, future researchers are encouraged to continue improving the system's security, scalability, and overall functionality to ensure that Eikonix remains responsive to the changing demands of digital commerce and the online art industry.

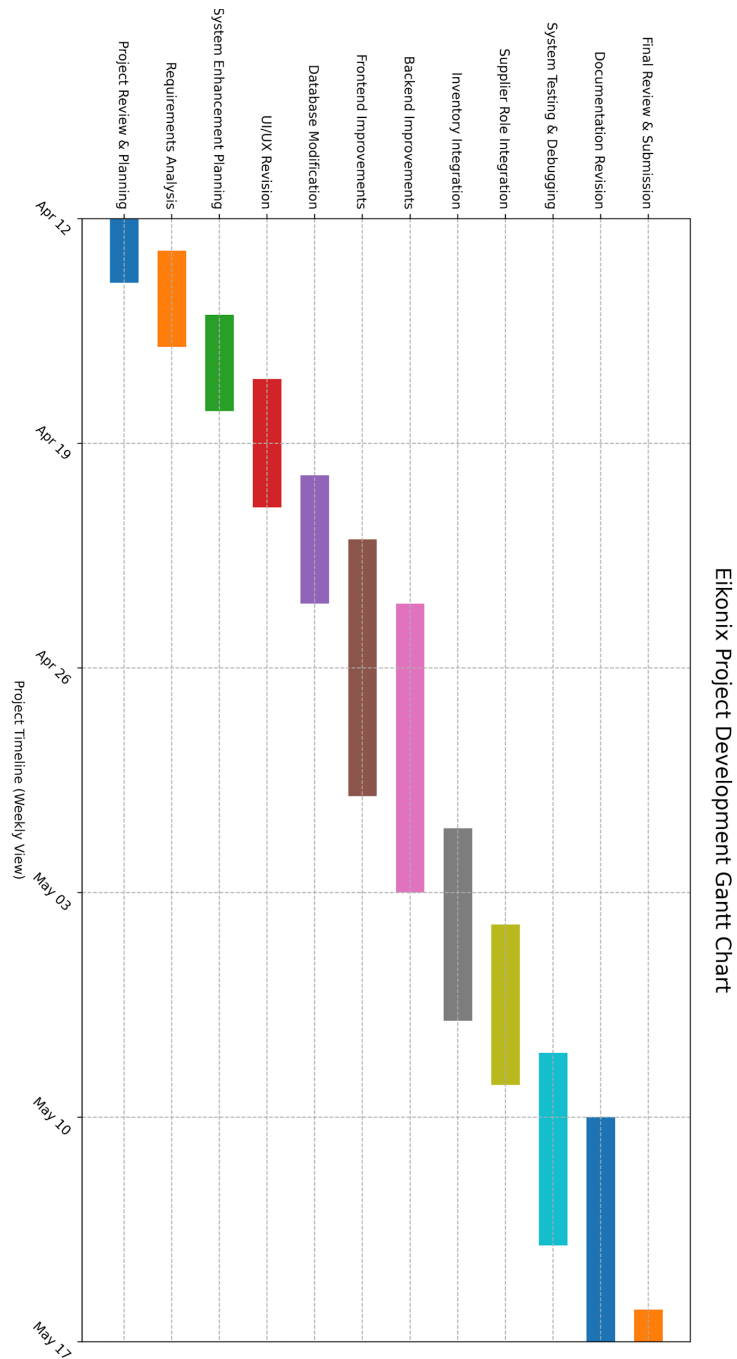


APPENDICES



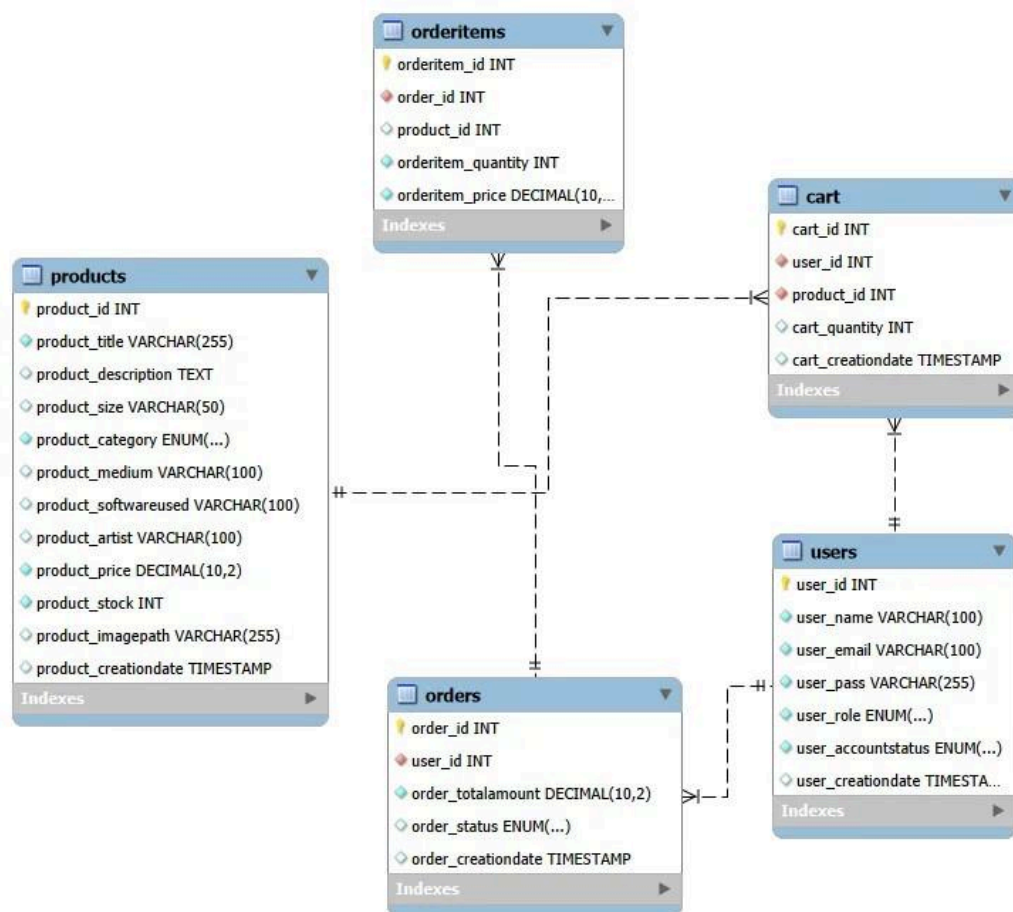
APPENDIX A

Gantt Chart





The Gantt chart outlines the systematic timeline and milestone distribution for the design, development, and implementation phases of the Eikonix platform. It visually tracks the progression of the development lifecycle, ensuring that tasks such as front-end adjustments and backend inventory integration are completed within schedule. This chronological roadmap allows the project team to manage resources efficiently and meet critical deadlines throughout the semester.

APPENDIX B**Database Schema**

The database schema provides the structural backbone of the Eikonix platform by defining the precise tables, exact data types, and structural keys needed for system operation. It establishes critical relationships among essential tables including users, products, orders, order items, and the cart to ensure rigid data integrity. Crucially, this schema supports the platform's core business logic by mapping out the product stock fields required to enforce the unique, single-artwork constraint.



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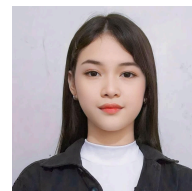


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- Capable of planning workflows and sketching simple UI blueprints.
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- Quick learner and open to acquiring new skills
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- Basic knowledge of computer applications and technology
- Fast learner and willing to learn new skills
- Good organizational and multitasking skills

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